Karman Vortex Shedding over a Circular Cylinder

Experiment 4 | 20ME406P Computational Fluid Dynamics and Heat Transfer Lab

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1. Aim and Scope

To resolve the alternating vortex shedding behind a circular cylinder using a D2Q9 BGK lattice Boltzmann solver and to estimate the Strouhal number from the lift history. The lift and drag coefficient histories, the density deviation, and the non-dimensional vorticity field are examined to decide whether the wake is steady or periodic.

The run analysed in this report is a user-defined (custom) case at Re = 47, which lies close to the critical Reynolds number at which a steady cylinder wake first becomes unstable to vortex shedding. The solver is a weakly compressible lattice-unit teaching model, so the results are interpreted as numerical trends and are not presented as validated high-fidelity CFD data.

2. Governing Equations

2.1 Macroscopic flow equations

At low Mach number the lattice Boltzmann method recovers the incompressible Navier-Stokes equations:

$$\nabla \cdot \mathbf{u} = 0 \tag{1}$$

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\frac{1}{\rho}\nabla p + \nu \nabla^2 \mathbf{u} \tag{2}$$

where u is the velocity vector, p the pressure, ρ the density and ν the kinematic viscosity. The flow is characterised by the Reynolds number based on the inlet speed U and the cylinder diameter D:

$$Re = \frac{UD}{\nu} \tag{3}$$

2.2 D2Q9 BGK lattice Boltzmann equation

The solver does not discretise Eqs. (1)-(2) directly. It evolves nine particle distribution functions fi along the discrete lattice velocities ci of the D2Q9 model, using a single-relaxation-time (BGK) collision operator:

$$f_i(\mathbf{x} + \mathbf{c}_i \Delta t,\, t + \Delta t) = f_i(\mathbf{x}, t) - \frac{1}{\tau}\left(f_i - f_i^{eq}\right) \tag{4}$$

The local equilibrium distribution is

$$f_i^{eq} = w_i \rho \left[1 + 3(\mathbf{c}_i \cdot \mathbf{u}) + \frac{9}{2}(\mathbf{c}_i \cdot \mathbf{u})^2 - \frac{3}{2}(\mathbf{u} \cdot \mathbf{u})\right] \tag{5}$$

The macroscopic density, momentum and pressure are recovered from the moments of fi:

$$\rho = \sum_{i=0}^{8} f_i, \quad \rho \mathbf{u} = \sum_{i=0}^{8} \mathbf{c}_i f_i, \quad p = c_s^2(\rho - 1) \tag{6}$$

The lattice speed of sound satisfies $c_s^2 = 1/3$ and the lattice viscosity is linked to the relaxation time τ through

$$\nu = c_s^2\left(\tau - \tfrac{1}{2}\right) \Rightarrow \tau = 0.5 + 3\nu \tag{7}$$

Here the lattice unit system is used ($\Delta x = \Delta t = 1$, $\rho 0 = 1$). A relaxation time close to 0.5 gives a very small viscosity and poor numerical stability, so the solver rejects $\tau \leq 0.505$.

2.3 Derived quantities

The hydrodynamic force on the cylinder is obtained by momentum exchange over all fluid-solid links, and is non-dimensionalised into drag and lift coefficients:

$$\mathbf{F} = \sum_{links} 2 f_i^*(\mathbf{x}_b)\, \mathbf{c}_i \tag{8}$$

$$C_d = \frac{F_x}{\frac{1}{2}\rho U^2 D}, \qquad C_l = \frac{F_y}{\frac{1}{2}\rho U^2 D} \tag{9}$$

The shedding frequency f is reported as a Strouhal number, and time is made dimensionless with the convective scale D/U:

$$St = \frac{fD}{U_\infty}, \qquad t^* = \frac{Ut}{D} \tag{10}$$

The non-dimensional vorticity used for the field plots is

$$\omega^* = \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right)\frac{D}{U} \tag{11}$$

3. Boundary Conditions

- **Inlet (left, x = 0):** Dirichlet condition. The equilibrium distribution for $\rho = 1$, $u = U = 0.10$ (lattice units) and $v = 0$ is imposed on the inlet column at every time step.
- **Outlet (right, x = Lx):** open, zero-gradient treatment. The distribution functions of the last column are copied from the neighbouring interior column.
- **Upper and lower boundaries (y = ±Ly/2):** periodic wrap-around in the cross-stream direction, which the solver records as a periodic far-field approximation. No wall is placed on these boundaries.
- **Cylinder surface:** no-slip condition enforced by bounce-back. Solid nodes are those satisfying $(i - ic)^2 + (j - jc)^2 \leq R^2$, and any population that would stream into a solid node is reflected back along its incoming link. Velocity inside the solid is set to zero.
- **Initial condition:** uniform density $\rho = 1$ and streamwise velocity $u = U$ everywhere in the fluid, with a small localised transverse perturbation $v = 0.05U \sin(2\pi j/ny) \exp[-((i -$

ic)/2D)²] near the cylinder to break the symmetry and allow shedding to start if the flow is unstable.

4. Numerical Method

4.1 Lattice and discrete velocities

A D2Q9 lattice (two dimensions, nine velocities) on a uniform Cartesian grid is used. The velocity set and weights are listed below, with $c = \Delta x/\Delta t = 1$.

Index i	Velocity ci	Weight wi
0	(0, 0)	4/9
1, 2, 3, 4	(1, 0), (0, 1), (−1, 0), (0, −1)	1/9
5, 6, 7, 8	(1, 1), (−1, 1), (−1, −1), (1, −1)	1/36

4.2 Solution algorithm

Each outer step of the solver consists of several lattice sub-steps. One lattice sub-step is carried out as follows:

1. Compute density and velocity from the distribution functions using Eq. (6); set the velocity to zero inside the cylinder.
2. Collision: relax every distribution towards its equilibrium, fi* = fi − (1/τ)(fi − fi^eq), using Eq. (5).
3. Streaming with bounce-back: shift each fi* by its lattice velocity ci. Populations whose destination node is solid are reversed instead, and the exchanged momentum 2fi*ci is accumulated into the force on the cylinder (Eq. 8).
4. Apply the inlet equilibrium condition and the zero-gradient outlet condition, and clear the populations inside the solid.

After the sub-steps of an outer step are completed, the drag and lift coefficients (force averaged over the sub-steps), the maximum speed, the minimum and maximum density, and the maximum density deviation $|\rho - 1|$ are stored. The run is aborted if any field becomes non-finite, if the maximum lattice speed exceeds 0.45, or if the density deviation exceeds 0.25. VTK field files are written every output_every outer steps and at the final step, together with the history CSV file and the summary file.

4.3 Strouhal-number estimation

The second half of the lift history is taken, its mean is removed, and it is multiplied by a Hann window before a fast Fourier transform is applied. The frequency of the largest spectral peak inside the admissible band $0.05 \le St \le 0.35$ is reported using Eq. (10). If the standard deviation of the lift signal is below 10^{-5}, the solver treats the signal as non-oscillatory and returns St = 0.

4.4 Case definition and parameters

The run analysed in this report corresponds to the following parameter set (RUN_MODE = custom):

Parameter	Variable	Value
Grid points, streamwise	nx	241
Grid points, cross-stream	ny	121
Reynolds number	Re = UD/ν	47
Outer steps	outer_steps	1200
Numerical iterations (input)	numerical_iterations	45 (4 lattice sub-steps per outer step)
Total lattice time steps	-	4800
Output interval	output_every	300 outer steps
Cylinder diameter	D	15 lattice units
Cylinder centre	(ic, jc)	(60, 60), i.e. x/D = 4, y/D = 0
Inlet speed	U	0.10 (lattice units)
Lattice viscosity	ν = UD/Re	0.03191
Relaxation time	τ	0.5957 (1/τ = 1.679)
Mach number	Ma = U/cs	≈ 0.173
Domain size (in cylinder diameters)	Lx × Ly	16D × 8D (x/D ∈ [0, 16], y/D ∈ [−4, 4])
Blockage ratio	D/ny	≈ 12.4 %
Final dimensionless time	Ut/D	32.0

Model scope: the solver is a weakly compressible lattice-unit teaching model with a staircase (bounce-back) representation of the circular boundary, only 15 lattice nodes across the diameter, and a short run of 32 convective time units. These restrictions should be kept in mind when the results are interpreted.

5. Results and Discussion

The figures below are the outputs of the Colab run and are discussed in.

5.1 Parameters

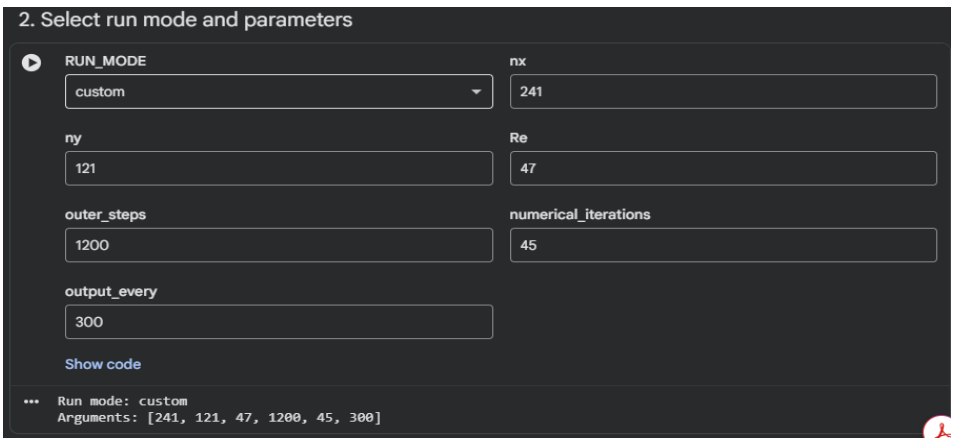


Figure 1. Run mode and parameter set used in the Colab notebook (RUN_MODE = custom, nx = 241, ny = 121, Re = 47, outer_steps = 1200, numerical_iterations = 45, output_every = 300).

5.2 Lift coefficient history

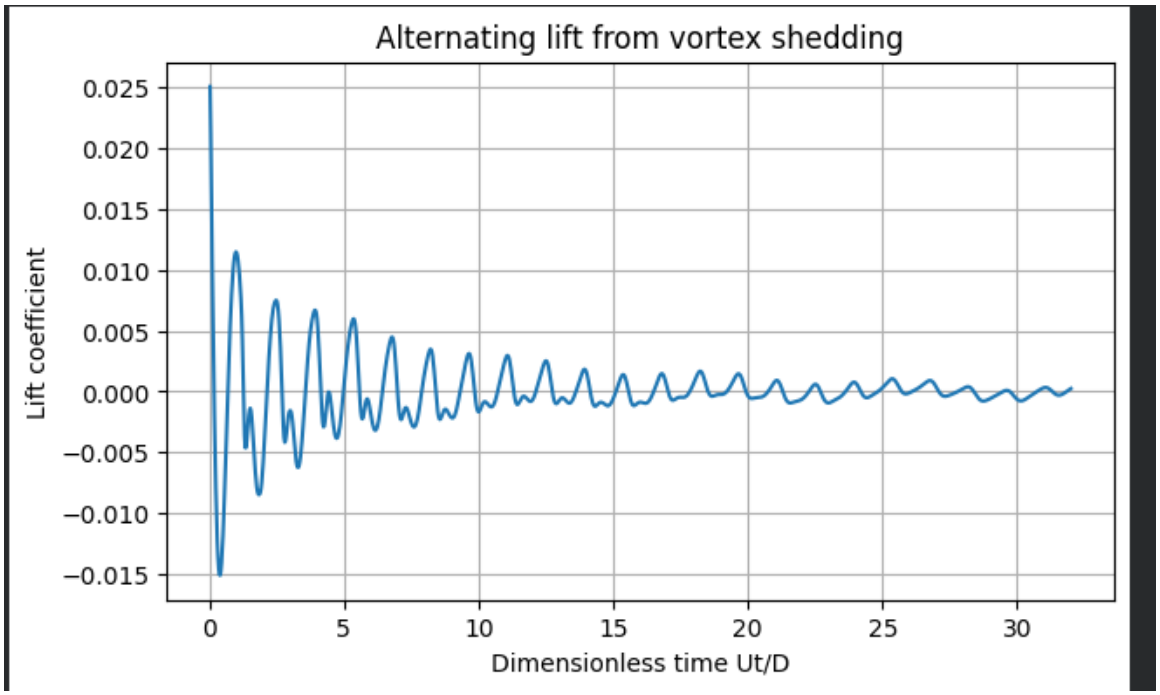


Figure 2. Lift coefficient history versus dimensionless time Ut/D.

- Figure 2 shows that the lift coefficient starts at about 0.025, drops to a minimum of about −0.015 within the first half convective time unit, and then oscillates about zero with a steadily shrinking envelope.
- The successive positive peaks fall from about 0.011 near Ut/D ≈ 1 to about 0.004 near Ut/D ≈ 7 and to roughly 0.001 by Ut/D ≈ 20, and by Ut/D = 32 the amplitude is only of the order of 4×10^{-4}. The mean lift stays at zero, as expected for a body symmetric about the flow direction, and the envelope never grows.
- A growing, finally saturating lift oscillation is the signature of the onset of vortex shedding. Here the opposite is observed: the disturbances introduced by the impulsive start and by the initial perturbation are damped out.
- The fast oscillations on the curve have a period of roughly 1.4 convective time units, which would correspond to St ≈ 0.7, several times larger than any shedding frequency expected for a cylinder at this Reynolds number.
- They are therefore most likely acoustic transients of the weakly compressible scheme, excited by the impulsive start, rather than vortex shedding. For comparison, a fully developed shedding cycle at Re = 100 gives a lift amplitude of order 0.3, about three orders of magnitude larger than the late-time amplitude seen here.

5.3 Drag coefficient history

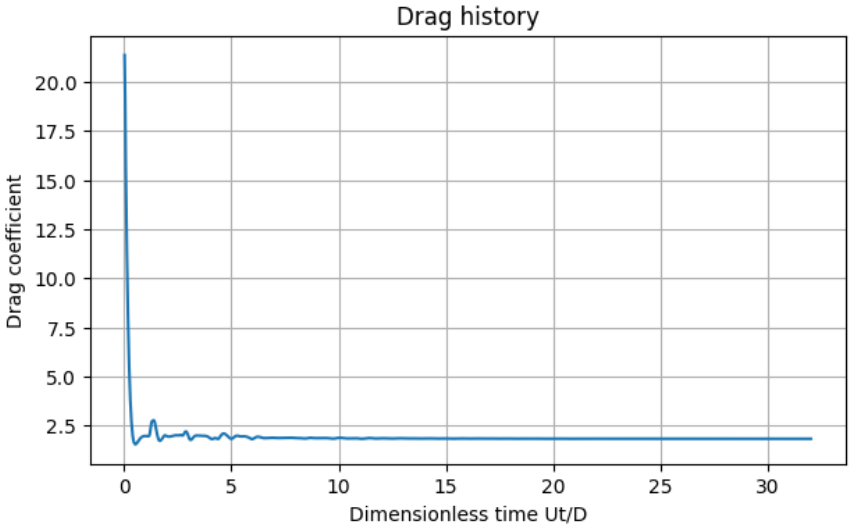


Figure 3. Drag coefficient history versus dimensionless time Ut/D.

- The drag coefficient in Figure 3 begins at a very large value of about 21 because the fluid is initialised at the free-stream speed and the no-slip cylinder is imposed instantaneously, which produces an impulsive pressure and momentum-exchange force.
- The drag collapses to about 1.5 within the first convective time unit, shows a small overshoot of about 2.7 near Ut/D ≈ 1.3 and minor ripples up to Ut/D ≈ 10, and then settles smoothly to a constant value of about 1.85 with no visible oscillation.
- A constant late-time drag together with a vanishing lift is characteristic of a steady wake. The steady drag coefficient of a cylinder at Reynolds numbers of 40-50 in unbounded flow is roughly 1.5-1.6, so the present value is somewhat higher.
- This offset is plausible for this teaching model: the blockage ratio is about 12 % with periodic upper and lower boundaries, only 15 lattice nodes span the diameter, and the stair-stepped bounce-back surface makes the effective body slightly larger than the ideal circle. The value should be quoted as a trend indicator and not as a validated drag coefficient.

5.4 Vorticity field and wake structure

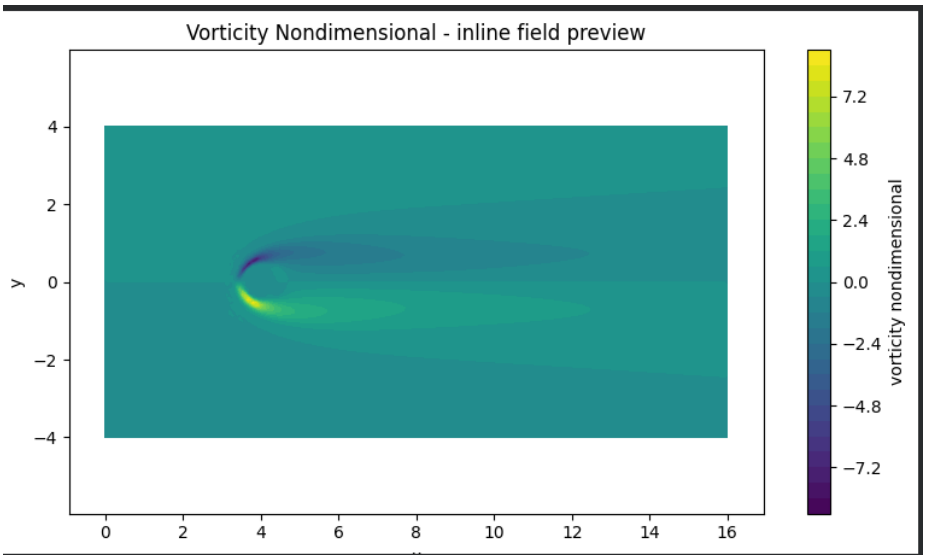


Figure 4. Non-dimensional vorticity field at the final step (Ut/D = 32), inline field preview of cylinder_vortex_final.vtk.

- The vorticity field in Figure 4 shows two thin shear layers of opposite sign attached to the flanks of the cylinder. The upper layer carries negative vorticity and the lower layer positive vorticity, consistent with the sign convention of Eq. (11) for flow in the +x direction, and the peak magnitude is about $|\omega^*| \approx 8$ close to the surface.
- Downstream, the two layers diffuse and form a long, almost symmetric wake that extends towards the outlet at x/D = 16. No rolled-up, alternating vortices of the Karman street are visible in the wake, so the flow at Ut/D = 32 is a steady, symmetric recirculating wake and not a periodic one. This agrees with the lift and drag histories.

6.4 Strouhal number and quantitative summary

Quantity	Value
Final dimensionless time Ut/D	32.0
Initial (impulsive-start) drag peak	≈ 21
Late-time mean drag coefficient	≈ 1.85 (steady)
Initial lift extremes	+0.025 to −0.015
Lift amplitude near Ut/D = 30	$\approx 4 \times 10^{-4}$ (decaying)
Estimated Strouhal number (solver output)	0.125
Maximum density deviation $\lvert\rho - 1\rvert$	0.191
Maximum lattice speed	0.171

- The solver reports St = 0.125 for this run. Because the lift signal is decaying and not periodic, this number is the location of the strongest spectral peak inside the search

band and must not be read as a physical shedding frequency. In addition, the analysed window is only the second half of the record, about 16 convective time units, so the frequency resolution of the transform is roughly $\Delta St \approx 1/16 \approx 0.06$. This is too coarse to resolve a Strouhal number to better than about ± 0.03 even when real shedding is present.

- For reference, the onset of shedding occurs near $Re \approx 47$ with St of about 0.12, and the laboratory case at $Re = 100$ is expected to give $St \approx 0.167$ within the range 0.14-0.20.
- The solver ran to completion without triggering its stability guards (maximum lattice speed above 0.45 or density deviation above 0.25), so the solution stayed bounded and finite.
- The density deviation of 0.191 and the maximum lattice speed of 0.171 should be compared with the Mach-number limit of the weakly compressible model ($Ma = 0.173$ at the inlet).

6.5 Interpretation near the critical Reynolds number

- $Re = 47$ is very close to the critical Reynolds number of about 46-47 for the first instability of the cylinder wake, so two effects are expected. First, the growth or decay rate of disturbances is close to zero, so a transient that dies out does so very slowly, which explains why the lift amplitude is still decreasing at $Ut/D = 32$ instead of vanishing quickly.
- Second, confinement by the finite domain tends to delay the onset of shedding slightly, so a stable wake at this Reynolds number is a physically reasonable outcome. The result is consistent with no sustained shedding at $Re = 47$ in this run, but the run is too short and too coarsely resolved to locate the critical Reynolds number itself.
- A run at $Re = 100$ (the laboratory case) is required to observe sustained periodic shedding and to verify the expected Strouhal number.

The main limitations affecting the interpretation are:

- short simulation time ($Ut/D = 32$), which limits the spectral resolution and the time available for slow transients to decay;
- periodic upper and lower boundaries combined with a blockage of about 12 %, which affect the drag level and the stability threshold;
- low resolution of the cylinder (15 nodes per diameter) with a staircase bounce-back surface;
- Weak compressibility ($Ma \approx 0.17$), which produces acoustic transients visible in the lift history.

7. Conclusions

- The D2Q9 BGK lattice Boltzmann solver with bounce-back and momentum-exchange forces ran stably for Re = 47 on a 241 × 121 grid for 4800 lattice time steps (Ut/D = 32), with no blow-up and with the stability guards untouched.
- The drag coefficient decays from an impulsive-start value of about 21 to a steady value of about 1.85, and the lift coefficient oscillates about zero with a continuously decaying amplitude of about 4×10^{-4} by the end of the run. No sustained oscillation develops.
- The vorticity field shows two attached shear layers of opposite sign and a long symmetric wake without alternating vortices, so the wake at Re = 47 is steady and symmetric within the simulated time, consistent with the flow being at or just below the onset of vortex shedding.
- The Strouhal number returned by the solver does not represent a physical shedding frequency for this run, since the lift signal is decaying and the spectral resolution of the 16-time-unit window is only about 0.06. A periodic wake and a reliable St (expected near 0.167) require the Re = 100 laboratory case.
- The results are qualitative trend indicators from a reduced teaching model, limited by the 12 % blockage, periodic cross-stream boundaries, coarse cylinder resolution and weak compressibility, and should not be presented as validated aerodynamic data.